



DiaTwin

A digital twin for Type 2 Diabetes that forecasts glucose, warns before spikes and lows, explains why, and simulates what-if choices

Digital Twin Challenge 2026 | Happiest Health | Team leader: Shivansh Porwal, IMT Ghaziabad

Diabetes care is reactive

101M

Indians living with diabetes

136M

more with prediabetes

11.4%

of the population has diabetes

What clinicians see today

- Lab values such as HbA1c, reviewed every few months
- Occasional glucose readings
- No view of what the next two hours hold
- Spikes, and lows on insulin or sulfonylureas, go unseen

Source: ICMR-INDIAB study, The Lancet Diabetes & Endocrinology, 2023

Meet DiaTwin



Fuse

Merge the static EHR with CGM, wearable and meal-diary streams for each patient.



Forecast

Glucose at +30, +60 and +120 minutes with an 80% uncertainty band.



Warn

Hyper and hypo alerts, with a why-this-forecast panel for the doctor.



Simulate

What if I eat, halve the meal, walk or skip? See the curve first.

Scope: one condition, Type 2 Diabetes, as the challenge requires. A proof of concept on synthetic data.

Built to the challenge brief

Challenge requirement	How DiaTwin meets it
Fuse static EHR with dynamic real-time data	Synthetic EHR + simulated CGM, heart rate, HRV, steps, sleep and meal diary: 43 fused features
Algorithmic model that predicts an adverse event	2-hour glucose forecast, spike-onset and hypoglycaemia alerts, with uncertainty
Conceptual UI dashboard for a doctor	Working Streamlit app: twin view, cohort board, what-if planner, explanations
Anonymised, open-source or synthetic data only	100% synthetic; Synthea-compatible importer; DPDP Act and HIPAA safe
Public GitHub repo with full README	README with every required item, MIT licence, tests, CI, Docker
Demo video, architecture diagram, presentation	20-minute script, 2-page architecture PDF, this deck

Two data streams, one patient



Static: simulated EHR

- Demographics, BMI, disease duration
- HbA1c, fasting glucose, eGFR, LDL
- TCF7L2 genetic risk marker
- Medication, comorbidities, lifestyle



Dynamic: wearables

- CGM glucose every 5 min, with dropouts
- Heart rate, HRV, steps, sleep stages
- Meal diary, only 85% logged
- Medication effects: dosing, skipped meals

400

virtual patients

14

days each

1.6M

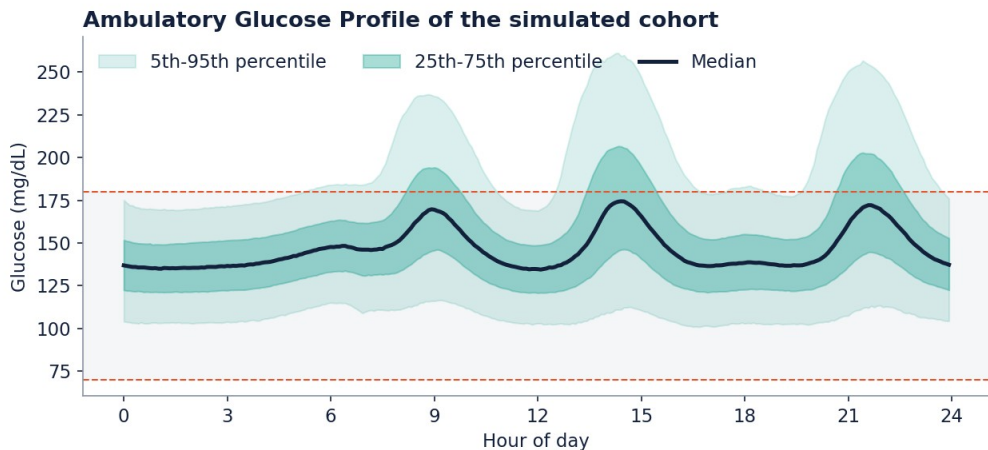
sensor readings

43

fused features

Synthetic only, as the sandbox rules require: DPDP Act and HIPAA safe by construction.

Is the synthetic data credible?



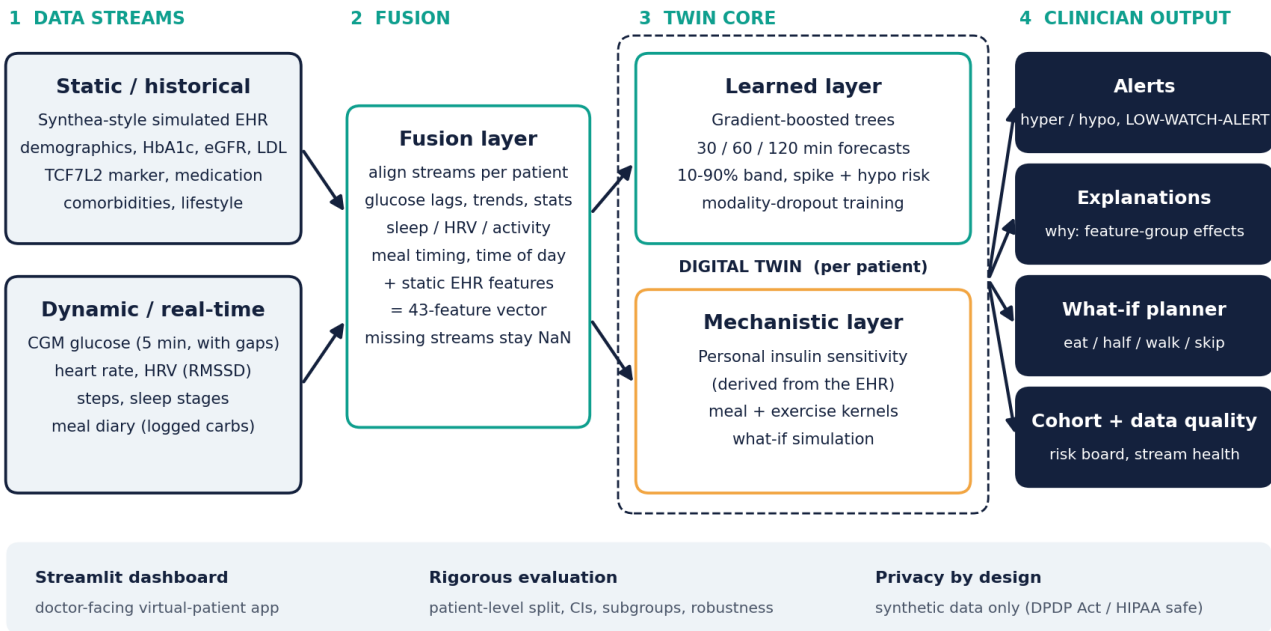
Metric	Cohort	Consensus
Time in range 70-180	86%	> 70%
Time above 180	14%	< 25%
Time below 70	0.12%	< 4%
GMI	6.9%	-
CV	17%	<= 36%

Honest caveat: the cohort is better controlled and less variable than typical real-world Indian Type 2 Diabetes, so absolute errors are optimistic. What transfers: model ranking, cold-start effect, robustness and the evaluation method.

System architecture

DiaTwin - System architecture

Type 2 Diabetes digital twin: fusion of static EHR and dynamic wearable/CGM streams

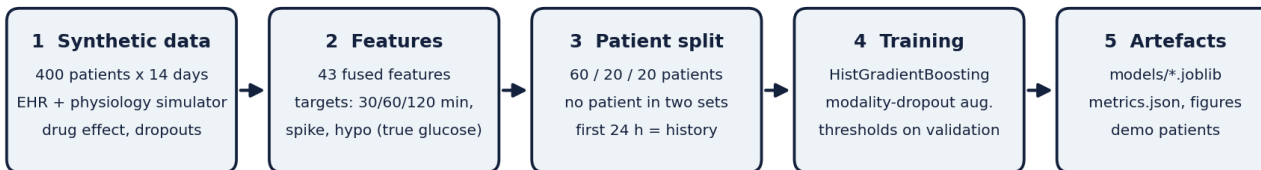


ML pipeline and evaluation

DiaTwin - ML pipeline and evaluation

Reproducible with one command: `python -m src.train`

BUILD



EVALUATION SUITE (all patient-level)



Serving: Streamlit dashboard loads the saved models and replays each virtual patient's stream. CI: GitHub Actions runs the test-suite on every push.

How the twin predicts

Learned layer

Gradient-boosted trees on 43 fused features: three forecast horizons, 10-90% band, spike and hypo classifiers. Modality-dropout training.

Mechanistic layer

Personal insulin sensitivity from the EHR drives meal and exercise kernels for what-if scenarios.

Explanation layer

Neutralise each data group to see how much it moves the forecast.

12.9

MAE in mg/dL at 2 h
persistence 26.5

0.93

ROC-AUC, spike alert
calibration ECE 0.009

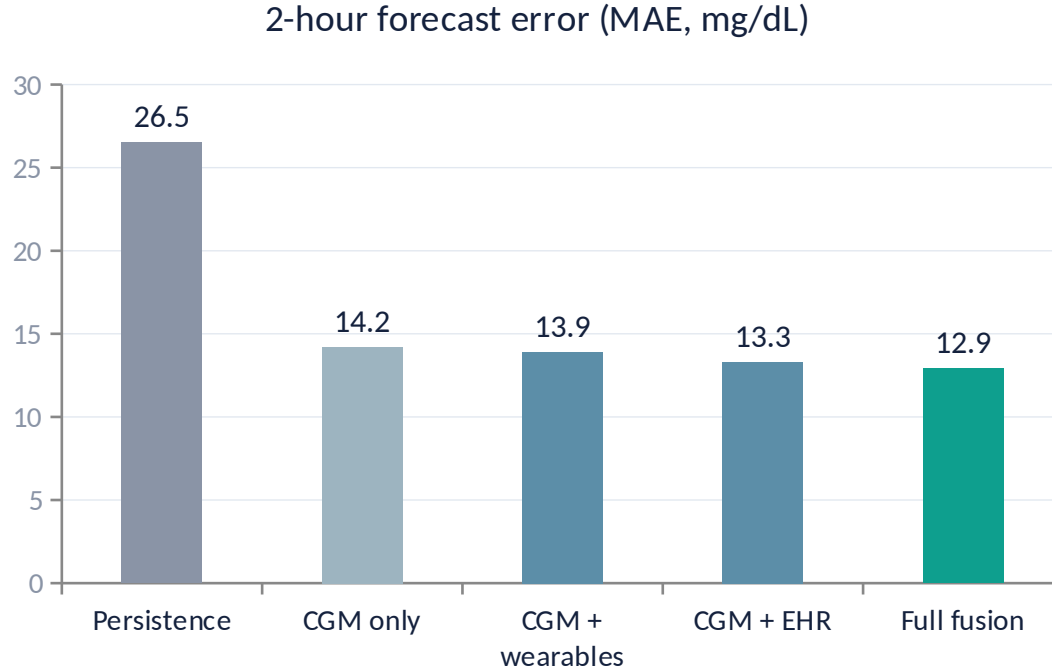
73%

within 15 mg/dL
of true glucose

79%

band coverage
nominal 80%

Does fusion help?



-51%

error vs persistence

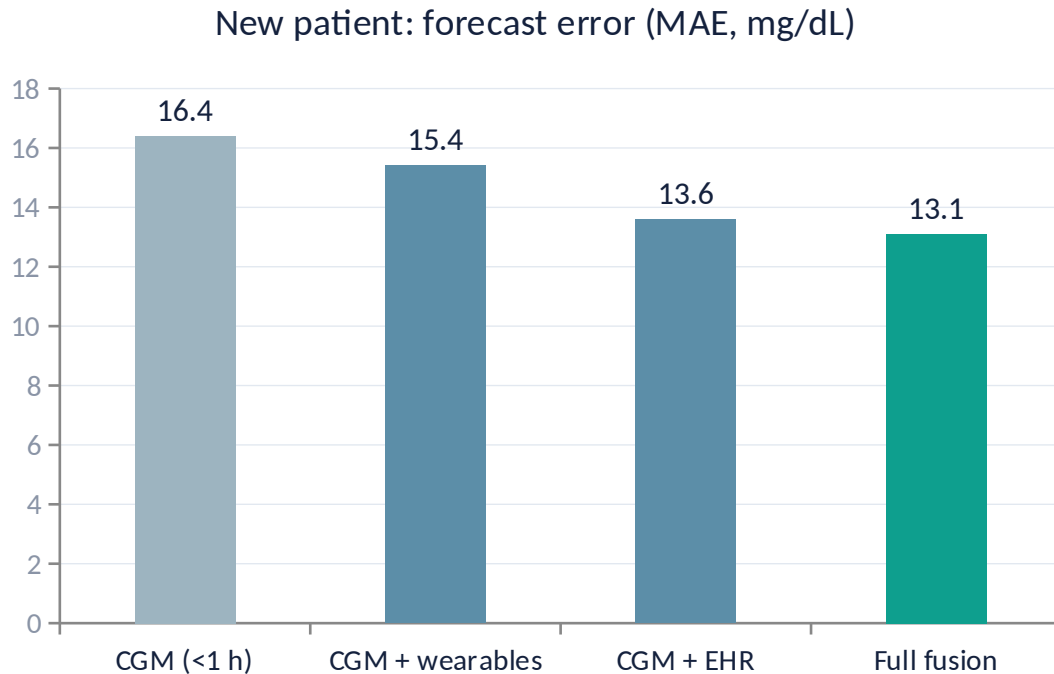
95% CI 12.4-13.5 mg/dL

1.22

mg/dL gain over CGM only

paired 95% CI 1.02 to 1.43

The cold-start advantage



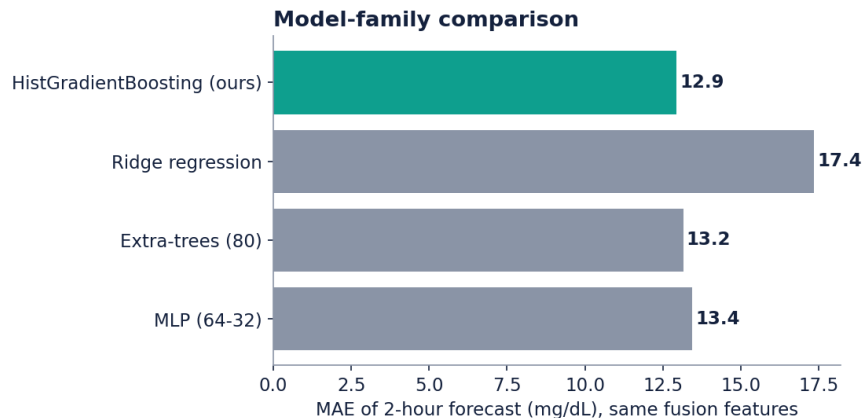
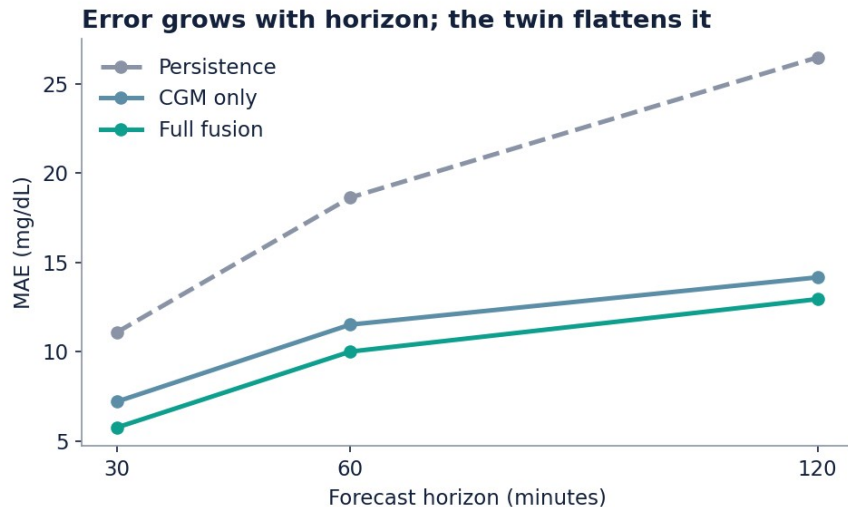
-20%

error for a new patient

16.4 to 13.1 mg/dL

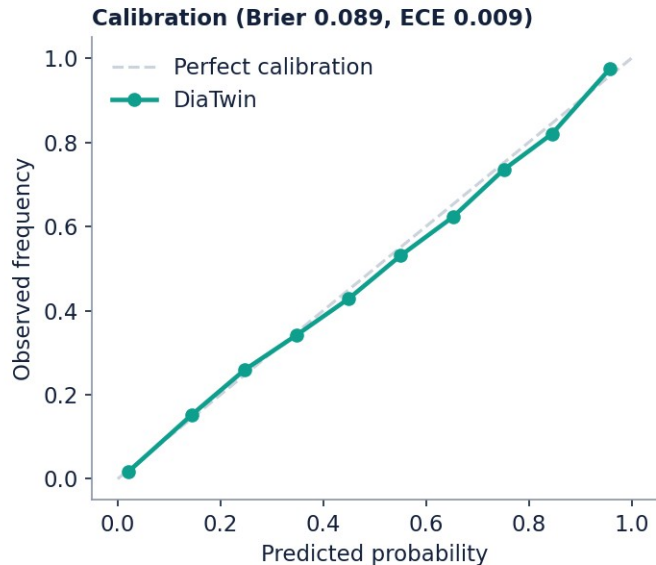
With under an hour of CGM data, the EHR supplies the baseline the sensor cannot yet show. That is what a twin is for: a head start on day one.

Horizons and model families



Error at +30 / +60 / +120 min: 5.7 / 10.0 / 12.9 mg/dL. Gradient boosting is the most accurate of four model families (extra-trees 13.2, MLP 13.4); ridge regression is far behind (17.4).

Alerts you can trust



0.929

spike-onset ROC-AUC

95% CI 0.923-0.937

+0.014

AUC gain vs CGM only

CI 0.010 to 0.020

0.089

Brier score

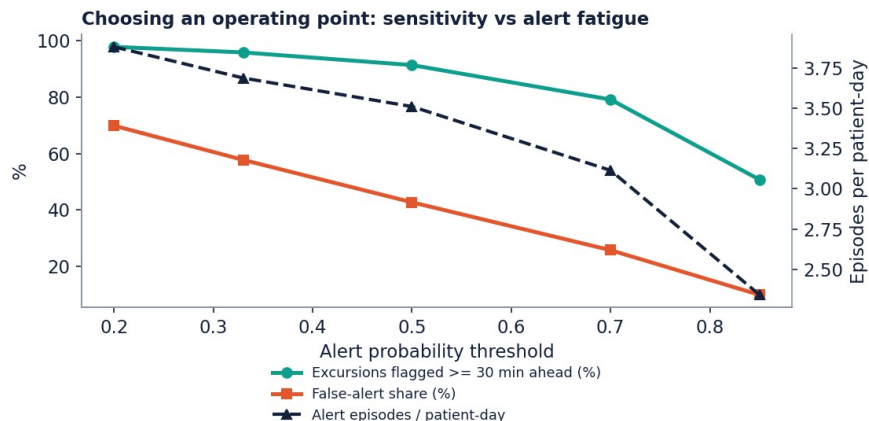
ECE 0.009

82% / 65%

recall / precision

at the tuned threshold

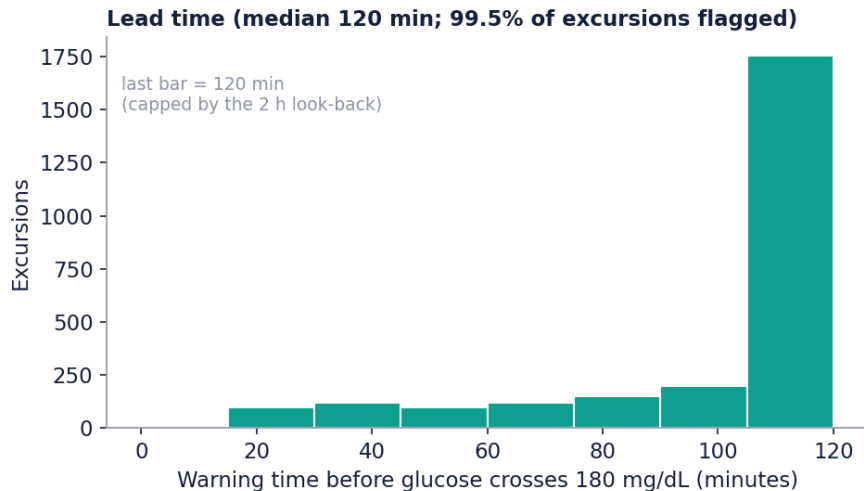
Alert fatigue is a design choice



Threshold	Alerts / day	False share	Flagged ≥ 30 min
0.20	3.9	70%	98%
0.33 *	3.7	58%	96%
0.50	3.5	43%	91%
0.70	3.1	26%	79%
0.85	2.3	10%	51%

At the tuned threshold (*), 58% of alert episodes are not followed by a spike. Raising the threshold cuts false alerts but misses more excursions. The clinician chooses the trade-off.

Early warning and hypoglycaemia

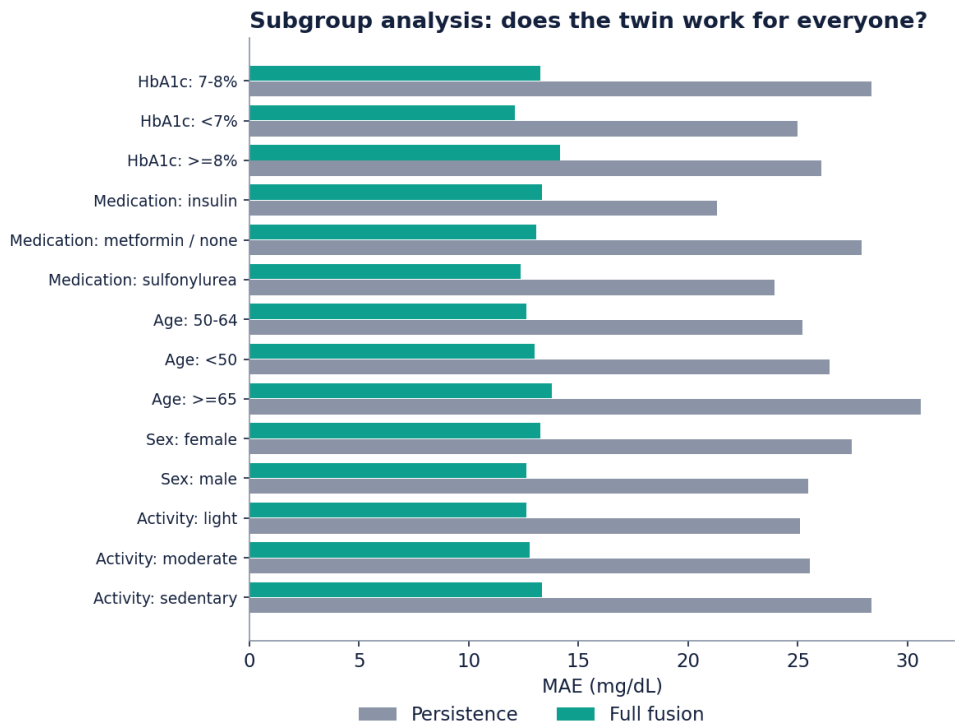


Spike warnings come a median 120 min ahead (capped at 120) and flag 99.5% of excursions.

Hypoglycaemia: a rare event

- 312 positive rows from 7 patients (6 on insulin)
- ROC-AUC 0.970 vs CGM only 0.973: no AUC advantage from the EHR
- Recall 89% at precision 9%: a screening aid, not a diagnosis

Does it work for everyone?



12.1-14.2

MAE range, mg/dL

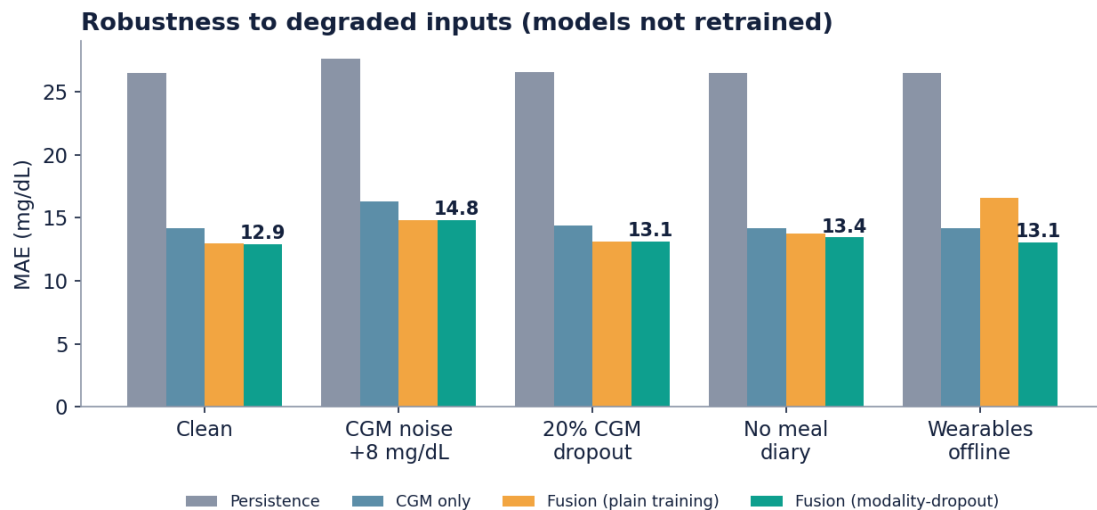
HbA1c <7% to HbA1c >=8%

>= 46%

better than persistence

in every subgroup of 10+ patients

Robust when inputs break



16.6

plain model, wearables off

worse than CGM only (14.2)

13.1

modality-dropout model

same scenario, same data

Holds up beyond the test set

Distribution shift

Unseen cohort of 120: older (63 vs 52 y), heavier (BMI 30.1 vs 27.7), longer disease (9.9 vs 6.6 y).

16.4

fusion MAE

CGM only 18.1, persistence 33.7

5-fold patient-level CV

Whole patients are held out in every fold.

13.43 + / - 0.06

fusion MAE

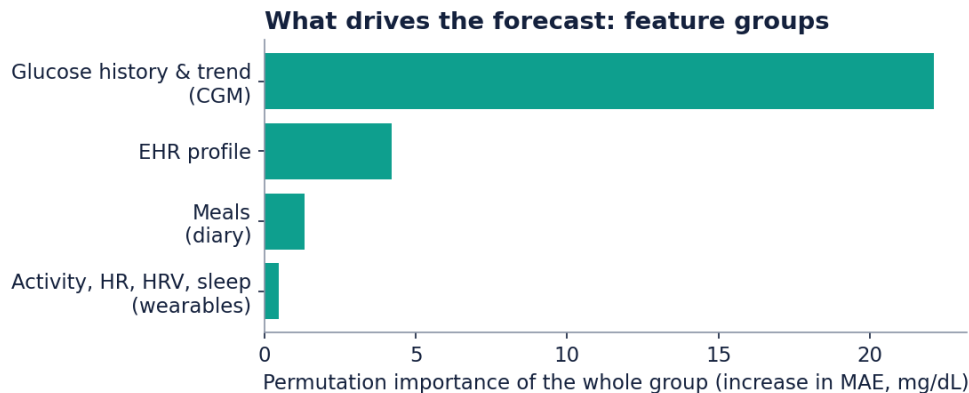
CGM only 14.60 +/- 0.14

13.5

cold-start fusion MAE

CGM-only cold 17.0

Why this forecast?

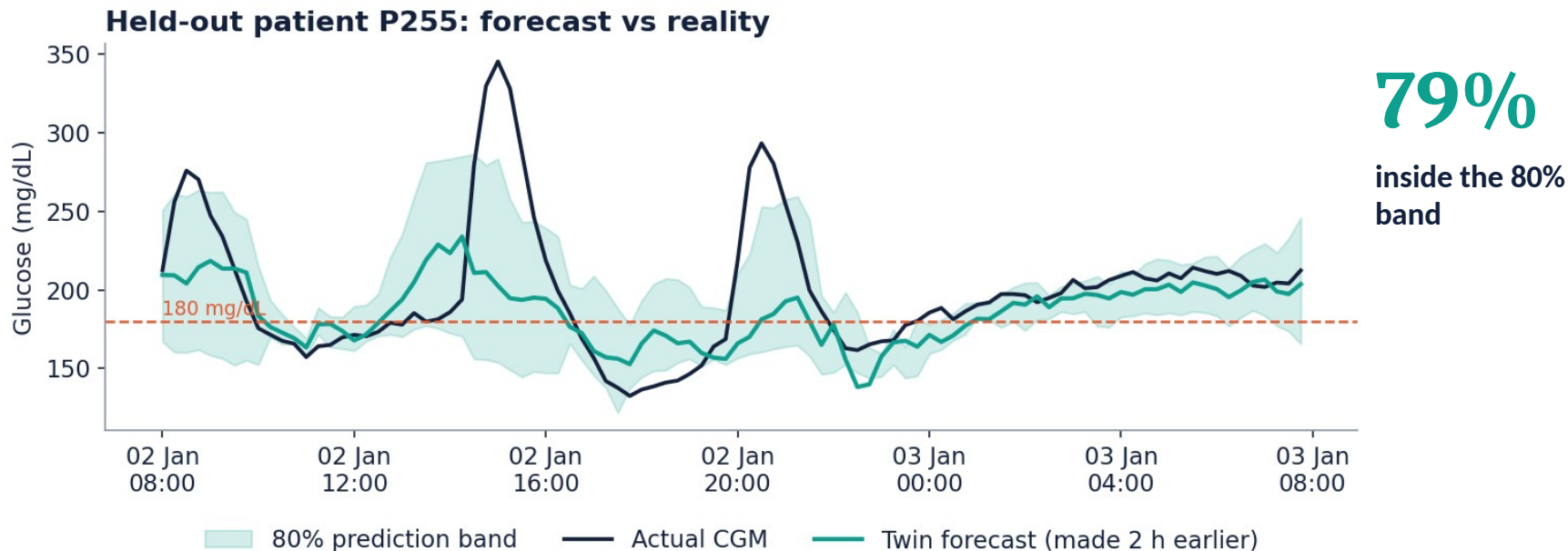


Globally, glucose history dominates, the EHR is second, meals third and wearables add least in this simulation, partly because exercise already shows up in the CGM trace.

In the dashboard

- Per-patient bars show how each data group pushes this forecast up or down
- Red raises glucose, green lowers it
- Stream-health warnings when data is missing

Forecast vs reality



Held-out virtual patient. Each forecast was made two hours earlier. The sharpest peaks are underestimated; the band widens to show uncertainty.

The doctor's view



Twin view

Forecasts, hyper and hypo tiles, why-this-forecast, clinical summary.



Cohort board

Every patient ranked by risk right now.



What-if planner

Eat, halve, walk after or skip: the next three hours.



Data fusion

EHR profile, wearable streams and the fused feature vector.



Data-quality simulator

Switch off wearables, diary or CGM and watch it degrade safely.



Model performance

Every evaluation in this deck, interactive.

Safe, private, accountable



Human in the loop

- Decision support only: it never doses
- Uncertainty is always shown
- Missing streams are flagged, not hidden



Privacy by design

- No real patient data
- Synthetic EHR and sensors
- DPDP Act and HIPAA safe



Fair and regulated

- Performance by 5 subgroup types
- Alert-fatigue trade-off exposed
- Software-as-medical-device review before any clinical use

Where DiaTwin fits

Common approaches today

- CGM apps: show glucose history and trends
- Care programmes: coaching and logging
- Hospital analytics: periodic risk scores

DiaTwin

- Fuses EHR and wearables per patient
- Forecasts, warns and explains
- Simulates choices before they are made

The gap we close: personalised, forward-looking, explainable and simulatable.

Our assessment of common approaches, not an exhaustive market survey.

Limits and roadmap

What we do not claim

- All results are on synthetic data: not clinical claims
- Simulated CGM is less variable than real life
- What-if layer shares the simulator's assumptions
- Hypoglycaemia: few events, 7 test patients
- Default alert threshold is noisy

Next steps

- Validate on real anonymised CGM and EHR data
- Learn the mechanistic parameters from data
- Add dose and medication timing; sequence models
- Clinician usability study, per-patient alert thresholds
- Drift monitoring and regulatory review

Predict. Warn. Simulate.

DiaTwin turns a patient's records and wearables into a virtual twin that a doctor can consult before the spike or the low, not after.

Team DiaTwin | IMT Ghaziabad | Code, README, technical report and demo video in the public GitHub repository